

Name : _____
Class : 12 MT _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL

1999 AP3

YEAR 12 HALF YEARLY HSC

MATHEMATICS

3/4 UNIT (COMMON)

Time allowed - 2 HOURS
(plus 5 minutes' reading time)

DIRECTIONS TO CANDIDATES:

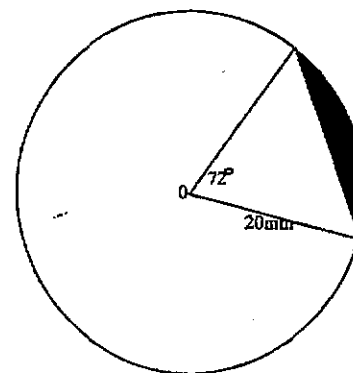
- * Attempt ALL questions.
- * All questions are of equal value
- * All necessary working should be shown in every question.
Full marks may not be awarded for careless or badly arranged work.
- * Standard Integrals are provided. Approved calculators may be used.
- * Each question attempted is to be returned on a new page clearly marked Question 1, Question 2, etc on the top of the page.

*Each page must show your class and your name.

QUESTION 1

Marks

- | | | |
|-----|---|---|
| (a) | Expand $(2x - y)^5$ | 2 |
| (b) | (i) Write down the expansion of $\cos(\alpha - \beta)$. | 3 |
| | (ii) Find the exact values of $\cos 45^\circ$ and $\cos 30^\circ$. | |
| | (iii) Hence find the exact value of $\cos 15^\circ$. | |
| (c) | (i) Convert 72° to radians, giving your answer in terms of π . | 3 |
| | (ii) Hence or otherwise, find the shaded area below correct to 3 significant figures. | |



- | | | |
|-----|--|---|
| (d) | Solve $\sin 2x = \sqrt{3} \cos 2x, 0 \leq x \leq 2\pi$. | 2 |
| (e) | Differentiate with respect to x | 2 |
| | (i) $\sqrt[3]{4x-1}$ | |
| | (ii) $\frac{x}{\cot x}$ | |

QUESTION 2

BEGIN A NEW PAGE

Marks

(a) Given $\int_0^3 f(t) dt = 6$, evaluate

4

(i) $\int_0^1 f(t) dt + \int_1^3 (f(t) + 1) dt$

(ii) $\int_3^0 (f(t) + t) dt$

(b) Find

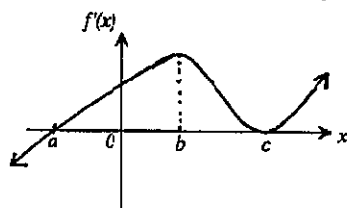
2

(i) $\int \frac{x^4 + 2x^3 + 3}{x^2} dx$

(ii) $\int \frac{dt}{(3-t)^2}$

(c) The gradient function of $y = f(x)$ is graphed below.

6



(i) Copy this diagram onto your answer sheet.

(ii) On the same diagram, sketch and label a possible graph of $y = f''(x)$.

(iii) State the domain where $y = f(x)$ is concave down.

(iv) Find the x values of any points of inflection.

(v) Find any stationary points and determine their nature.

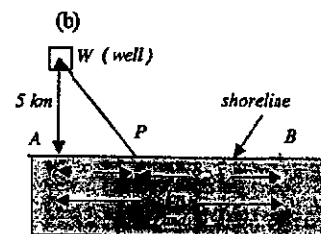
QUESTION 3

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Marks

(a) Find the equation of any asymptotes of the curve $y = \frac{x^2 + x + 1}{x}$.

2



An offshore oil well is located at a point W , which is 5 km from the closest shorepoint A on a straight shoreline.

The oil is to be piped to a shorepoint B that is 8 km from A by piping it on a straight line under water from W to some shorepoint P between A and B and then on to B via a pipe along the shoreline.

If the cost of laying the pipe is \$125 000 per km under water and \$75 000 per km over land.

Let x km be the distance between A and P and C (in thousands of dollars) be the cost for the entire pipeline.

(i) Show that the cost is given by $C = 125\sqrt{x^2 + 25} + 75(8 - x)$

3

(ii) Find the domain for x

1

(iii) Find where the point P should be located to minimise the cost of laying the pipe?

6

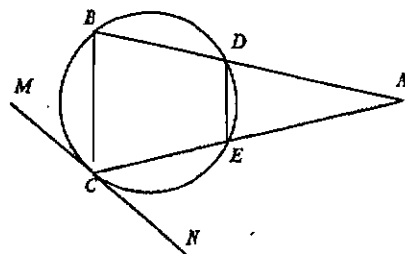
QUESTION 4

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Marks

(a)

6



ABC is a triangle in which $AB = AC$. A circle through B and C cuts AB at D and AC at E . MCN is the tangent at C to the circle through B, C, E, D .

(i) Copy the diagram onto your answer sheet.

(ii) Show that $DE \parallel BC$.

(iii) Show that $\angle ACN = \angle BCD$.

(b) $P(2at, at^2)$ is a variable point on the parabola $x^2 = 4ay$, whose focus is S .

6

$Q(x, y)$ divides the interval from P to S in the ratio $t^2 : 1$.

(i) Find x and y in terms of a and t .

(ii) Verify that $\frac{y}{x} = t$.

(iii) Prove that as P moves on the parabola, Q moves on a circle, and state its centre and radius.

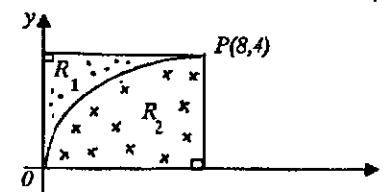
QUESTION 5

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Marks

(a)

7



OP is an arc of the curve $y^3 = x^2$.

Calculate the volume of the solids generated when

(i) Region R_1 revolves around the y -axis

(ii) Region R_2 revolves around the x -axis

(iii) Region R_2 revolves about the y -axis.

(b) (i) Express $\sin x - \cos x$ in the form $A \sin(x - \alpha)$ with $A > 0$ and $0 < \alpha < \frac{\pi}{2}$.

5

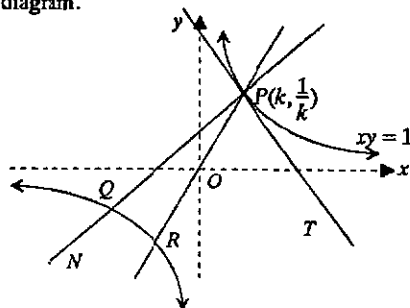
(ii) Determine $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}}$.

QUESTION 6

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Marks

$P(k, \frac{1}{k})$ is a point on the curve $xy = 1$ where k is a real number, $k \neq 0$.
 PT is the tangent to the curve at P and PN is the normal at P .
 POR is the line passing through P , the Origin O and R as shown on the diagram.



- (a) Find the equation of the line passing through O and P . 1
- (b) The line in part (a) intersects the curve again at R . Find the coordinates of R . 2
- (c) Show that the equation of the tangent at P is given by:

$$x + k^2y = 2k.$$
 2
- (d) Find the equation of the normal line at P . 2
- (e) Show that when the normal intersects the curve again at Q , the equation formed to solve is the quadratic equation given by:

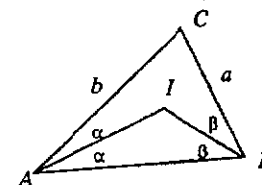
$$k^3x^2 - (k^4 - 1)x - k = 0.$$
 Hence find the coordinates of point Q . 3
- (f) Show that: $QR \perp PR$. 2

QUESTION 7

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Marks

- (a) Given the polynomial function $P(x) = x^3 - 2x^2 - 6x + 4$, when $P(x) = 0$, $P(x)$ has one rational root and two irrational roots.
 - (i) Find the rational root of $P(x) = 0$. 1
 - (ii) Without finding the irrational roots of $P(x)$, show that one of the irrational roots of this equation lies between $x = 3$ and $x = 4$. 1
 - (iii) Using $x = 3.5$ as a first approximation, apply Newton's Method once to find a better approximation to the root, to 2 decimal places. 3
 - (iv) Sketch $P(x) = x^3 - 2x^2 - 6x + 4$. 1
 - (v) Explain why $x = 2$ would not be a good approximation to use when solving $P(x) = 0$ using Newton's Method. 1
 - (vi) Find the area bounded by the curve, $x = -3$, $x = -2$ and the x -axis. 2
- (b) IA and IB bisect angles CAB and CBA as shown in the diagram below. 3



Prove that $\frac{IB}{IA} = \frac{a \cos \beta}{b \cos \alpha}$.

Question 1

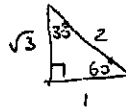
$$\begin{aligned}
 (a) (2x-y)^5 &= 1(2x)^5(-y)^0 + 5(2x)^4(-y)^1 + 10(2x)^3(-y)^2 + 10(2x)^2(-y)^3 + 5(2x)^1(-y)^4 + (2x)^0(-y)^5 \\
 &= 32x^5 - 80x^4y + 80x^3y^2 - 40x^2y^3 + 10xy^4 - y^5
 \end{aligned}$$

①

$$(b) (i) \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta \quad ①$$

$$(ii) \cos 45^\circ = \frac{1}{\sqrt{2}}$$

$$\cos 30^\circ = \frac{\sqrt{3}}{2}$$

① need both

$$\begin{aligned}
 (iii) \cos 15^\circ &= \cos(45^\circ - 30^\circ) \\
 &= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ \\
 &= \frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} \\
 &= \frac{\sqrt{3} + 1}{2\sqrt{2}} \quad ① \\
 &= \frac{\sqrt{6} + \sqrt{2}}{4}
 \end{aligned}$$

3

$$(c) (i) 1^\circ = \frac{\pi}{180}$$

$$72^\circ = \frac{72\pi}{180}$$

$$= \frac{8\pi}{20}$$

$$= \frac{2\pi}{5} \quad ①$$

$$\begin{aligned}
 (ii) \text{Area of minor segment} &= \frac{1}{2} r^2 (\theta - \sin \theta) \\
 &= \frac{1}{2} \cdot 20^2 \cdot \left(\frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \quad ① \\
 &= 61.11610 \dots \\
 &= 61.1 \text{ mm}^2 \text{ (3 sig. fig.)} \quad ①
 \end{aligned}$$

3

$$(d) \sin 2x = \sqrt{3} \cos 2x$$

$$\frac{\sin 2x}{\cos 2x} = \sqrt{3}$$

$$\tan 2x = \sqrt{3}$$

$$2x = \frac{\pi}{3}, \pi + \frac{\pi}{3}, 2\pi + \frac{\pi}{3}, 3\pi + \frac{\pi}{3}, \dots \quad ①$$

$$x = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6}, \dots \quad ①$$

2

$$\begin{aligned}
 (e) (i) \frac{d}{dx} \sqrt[3]{4x-1} &= \frac{d}{dx} (4x-1)^{\frac{1}{3}} \\
 &= \frac{1}{3} \cdot (4x-1)^{-\frac{2}{3}} \cdot 4 \\
 &= \frac{4}{3} (4x-1)^{-\frac{2}{3}} \text{ or } \frac{4}{3 \sqrt[3]{4x-1}^2} \quad ①
 \end{aligned}$$

$$\begin{aligned}
 (ii) \frac{d}{dx} \left(\frac{x}{\cot x} \right) &= \frac{d}{dx} x \tan x \\
 &= x \cdot \sec^2 x + \tan x \cdot 1 \\
 &= x \sec^2 x + \tan x \quad ①
 \end{aligned}$$

2

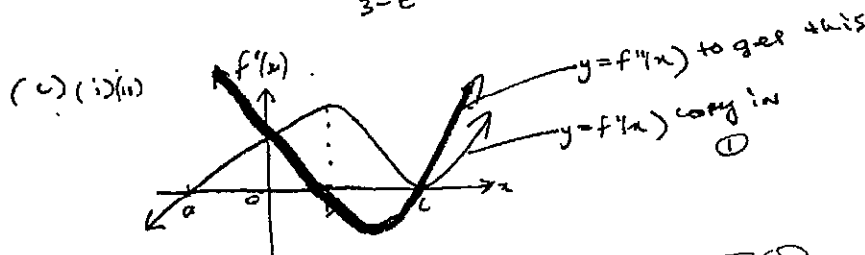
Question 2

$$\begin{aligned}
 (a) (i) &= \int_0^1 f(t) dt + \int_1^3 f(t) dt + \int_3^4 1 dt \\
 &= \int_0^3 f(t) dt + [t]_1^3 \\
 &= 6 + 3 - 1 \\
 &= 8
 \end{aligned}$$

$$\begin{aligned}
 (ii) &= \int_3^0 f(t) dt + \int_3^0 t dt \\
 &= - \int_0^3 f(t) dt + \left(\frac{t^2}{2} \right)_3^0 \\
 &= -6 + 0 - \frac{9}{2} \\
 &= -10\frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 (b) (i) & \int \frac{x^4 + 2x^3 + 3}{x^2} dx \\
 &= \int (x^2 + 2x + 3x^{-2}) dx \\
 &= \frac{x^3}{3} + x^2 - 3x^{-1} + C \\
 &= \frac{x^3}{3} + x^2 - \frac{3}{x} + C
 \end{aligned}$$

$$\begin{aligned}
 (ii) & \int \frac{dt}{(3-t)^2} = \int (3-t)^{-2} dt \\
 &= \frac{(3-t)^{-1}}{-1 \times -1} \\
 &= \frac{1}{3-t} + C
 \end{aligned}$$



(iii) $b < x < c$

(iv) $x = b$

x	b^-	b	b^+
$f'(x)$	+	0	-

There is a change in concavity at $x=b$

x	c^-	c	c^+
$f''(x)$	-	0	+

There is a change in concavity at $x=c$

$\therefore$ Point of inflection at $x=b, x=c$

(i) $f'(a) = 0$

$f''(a) > 0$

$\therefore x=a$ is a rel. min. turning pt

Tests

$f'(x) > 0$

x	c^-	c	c^+
$f'(x)$	+	0	+
Slope	—	—	—

$\therefore x=c$ is a stationary point of inflection H.P.O.I

- (i) 1
- (ii) 1
- (iii) 1
- (iv) 1+1

(v) 1+1

Question 3

(a) $y = \frac{x^2 + x + 1}{x}$

vertical asymptote $x=0$

diagonal asymptote $y=x+1$

$y = x + 1 + \frac{1}{x}$

2

(b) $WP^2 = 5^2 + x^2$

$WP = \sqrt{25 + x^2}$

Cost = $125000 \times WP + 75000 \times PB$

$C = 125\sqrt{25+x^2} + 75(8-x)$ (C is in thousands of dollars)

3

(i) $x^2 + 25 > 0$ and $8-x \geq 0$ and $x \geq 0$

$x^2 > -25$

$-x \geq -8$

$x \leq 8$

$\therefore x \in \mathbb{R}$

$\therefore 0 \leq x \leq 8$

1

1

(ii) $\frac{dC}{dx} = 125 \cdot \frac{1}{2} (25+x^2)^{-\frac{1}{2}} \cdot 2x - 75$

$= \frac{125x}{\sqrt{25+x^2}} - 75$

$\frac{dC}{dx} = 0$ for max/min

$\frac{125x}{\sqrt{25+x^2}} - 75 = 0$

$125x = 75\sqrt{25+x^2}$

$5x = 3\sqrt{25+x^2}$

Square both sides

$25x^2 = 9(25+x^2)$

$25x^2 = 225 + 9x^2$

$16x^2 = 225$

$x^2 = \frac{225}{16}$

$x = \pm \frac{15}{4}$

Since $0 \leq x \leq 8$

$x = \frac{15}{4}$

(1) This allocated at end of question after testing

	3	4
x	$\frac{15}{4}$	$\frac{15}{4}$
$\frac{dC}{dx}$	-	+
Sign	-	+

$\therefore$ rel. min when $x = \frac{15}{4}$

test end points of $0 \leq x \leq 8$

$x=0$

Cost = $(5 \times 125 + 8 \times 75) \times 1000$

$= \$1225000$

$x=8$

$WP = \sqrt{8^2 + 5^2}$

$= \sqrt{89}$

$PB = 0$

$\therefore$ Cost = $(\sqrt{89} \times 125) \times 1000$

$= \$1179247.642...$

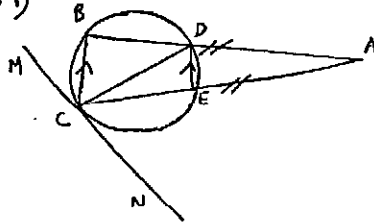
$= \$1179247.64$

$\therefore$ The point P should be located $\frac{15}{4} = 3.75 = 3.75 \text{ km}$

1

Question 4

(a) i)



ii) $\triangle ABC$ is isosceles ($AB = AC$)

Let $\angle ABC = x$

$\therefore \angle ABC = \angle BCA = x$ (equal base $\angle$'s, isos Δ) (1)

$$\angle BDE = 180^\circ - \angle ACB \quad (\text{opp } \angle\text{'s} \text{ ychase \& wad are supp})$$

$$= 180^\circ - \alpha \quad (1)$$
$$\angle CAD + \angle BDE = 180^\circ \quad (1)$$

and these are wintering notes ①

∴ 30(1) DE

iii) $\angle ACN = \angle CDE$ (angle in the alternate segment theorem) (1)

$$\angle CDE = \angle BCD \quad (\text{alt- } \angle\text{'s } \Rightarrow BC \parallel DE) \quad (1)$$
$$\therefore \angle ACN = \angle BCD$$

(b)

A graph of a parabola $x^2 = 4ay$ in the xy -plane. The origin is labeled $(0,0)$. A point $P(2at, at^2)$ is marked on the parabola. A line segment connects the origin to P , and a tangent line is drawn from P to the y -axis. The distance from the origin to P is labeled S .

$$\begin{aligned} (i) \quad \bar{Q} &= \left(\frac{nx_1 + m\bar{x}_2}{n+m}, \frac{ny_1 + my_2}{n+m} \right) \\ &= \left(\frac{1 \times 2at^2 + t^2 \times 0}{t^2 + 1}, \frac{1 \times at^2 + t^2 \times a}{t^2 + 1} \right) \\ &= \left(\frac{2at^2}{t^2 + 1}, \frac{2at^2}{t^2 + 1} \right) \end{aligned}$$

$$(11) \frac{y}{x} = \frac{\frac{2x+2}{t^2+1}}{\frac{2xt}{t^2+1}} = \frac{2x+2}{2xt}$$

$$\therefore \frac{y}{x} = b \quad (1)$$

$$\begin{aligned}(m) \quad x &= \frac{2xy}{x^2+1} \\ &= \frac{2a\left(\frac{y}{x}\right)}{\left(\frac{y}{x}\right)^2+1} \\ &= \frac{\frac{2ay}{x}}{\frac{y^2+x^2}{x^2}}\end{aligned}$$

$$\therefore x = \frac{2xy}{x^2 + y^2} \quad (1)$$

$$3x^2 + y^2 = 2xy$$

$$x^2 + y^2 - 2ay = 0$$

$$x^2 + (y-a)^2 = a^2 \quad (1)$$

create active (0, -)

admission a unit

30

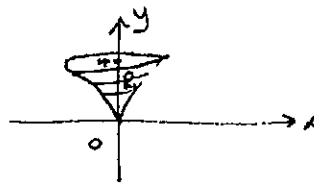
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Question 5

$$(a) \quad (i) \text{ Volume} = \pi \int_0^4 y^3 dx$$

$$= \pi \left[\frac{y^4}{4} \right]_0^4 \quad (1)$$

$$= \pi \left[\frac{4^4}{4} - 0 \right]$$



$$= 64\pi \text{ cubic units} \quad (1)$$

$$(ii) \text{ Volume} = \pi \int_0^8 x^{4/3} dx \quad (1)$$

$$= \pi \left[\frac{3x^{7/3}}{7} \right]_0^8 \quad (1)$$

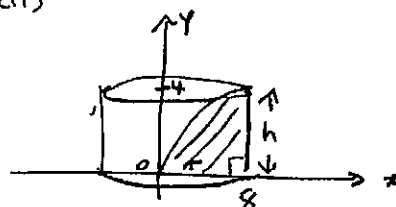
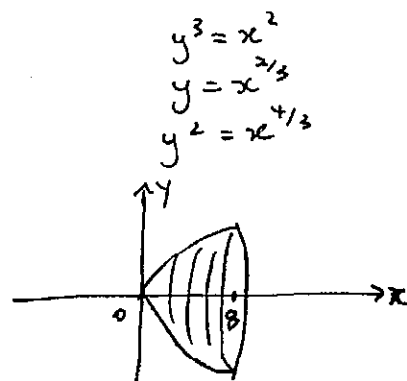
$$= \pi \left[\frac{3}{7} 8^{7/3} - 0 \right]$$

$$= \pi \cdot \frac{3}{7} \cdot 2^7$$

$$= \frac{384}{7} \pi \text{ cubic units} \quad (1)$$

$$(iii) \text{ Volume} = \pi \times 8^2 \times 4 - 64\pi \quad (1) \quad \leftarrow \text{Qs (i)}$$

$$= 192\pi \text{ cubic units} \quad (1)$$



$$(b) \quad (i) \sin x - \cos x = A \sin(x - \alpha)$$

$$= A \sin x \cos \alpha - A \cos x \sin \alpha \quad (1)$$

$$\therefore A \cos \alpha = 1 \quad \dots \dots \dots (1)$$

$$A \sin \alpha = 1 \quad \dots \dots \dots (2)$$

$$(1)^2 + (2)^2$$

$$A^2 \sin^2 \alpha + A^2 \cos^2 \alpha = 1 + 1$$

$$A^2 (\sin^2 \alpha + \cos^2 \alpha) = 2$$

$$A^2 = 2$$

$$A = \sqrt{2} \quad (1)$$

$$\text{From (1), (2)} \quad \left. \begin{array}{l} \cos \alpha = \frac{1}{\sqrt{2}} \\ \sin \alpha = \frac{1}{\sqrt{2}} \end{array} \right\} \text{quad I, } \triangle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1$$

$$\therefore \alpha = \frac{\pi}{4}$$

$$\sin x - \cos x = \sqrt{2} \sin \left(x - \frac{\pi}{4} \right)$$

$$(ii) \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \cos x}{x - \frac{\pi}{4}} = \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{2} \sin \left(x - \frac{\pi}{4} \right)}{x - \frac{\pi}{4}} \quad (1)$$

$$= \sqrt{2} \quad (1)$$

7

5

QUESTIONS

$$(a) m_{op} = \frac{\frac{1}{k} - 0}{k - 0}$$

$$= \frac{1}{k^2}$$

eqn of is.

$$y = \frac{1}{k^2}x$$

$$x - k^2y = 0$$

①

1

(b) Solve simultaneously $y = \frac{1}{k^2}x$

$$\text{and } y = \frac{1}{x}$$

$$\therefore \frac{1}{k^2}x = \frac{1}{x}$$

$$x^2 = k^2$$

$$x = k \text{ or } x = -k$$

Since P is $(k, \frac{1}{k})$

$$R \text{ is } (-k, -\frac{1}{k})$$

①

2

$$(c) y = x^{-1}$$

$$\frac{dy}{dx} = -x^{-2}$$

$$= -\frac{1}{x^2}$$

$$\text{at } x = k, \frac{dy}{dx} = -\frac{1}{k^2}$$

$$\therefore \text{gradient of tangent} = -\frac{1}{k^2}$$

$$\text{equation of tangent is } y - \frac{1}{k} = -\frac{1}{k^2}(x - k)$$

$$k^2y - k = -x + k$$

$$x + k^2y = 2k \text{ as required}$$

①

$$(d) \text{Slope of tangent} = -\frac{1}{k^2}$$

$$\therefore \text{slope of normal} = k^2 \quad (m_1 m_2 = -1)$$

①

$$\text{Equation of normal } y - \frac{1}{k} = k^2(x - k)$$

$$ky - 1 = k^3x - k^4$$

$$k^3x - ky = k^4 - 1$$

①

$$y = k^2x + \frac{1}{k} - k^3$$

2

(e) Solve simultaneously

$$y = \frac{1}{x} \text{ and}$$

$$k^3x - ky = k^4 - 1$$

eqn ①

eqn ②

Sub eqn ① into eqn ②

$$k^3x - k\left(\frac{1}{x}\right) = k^4 - 1$$

$$k^3x^2 - k = (k^4 - 1)x$$

$$k^3x^2 - (k^4 - 1)x - k = 0$$

}

①

Since $P(k, \frac{1}{k})$ lies on the normal

$x=k$ is a root of the equation

or use $\alpha\beta = \frac{k^4}{k^3}$

method ①

$$\begin{array}{r} k^3x + 1 \\ x-k \overline{) K^3x^2 - (K^4-1)x - K} \\ \underline{K^3x - K^4x} \\ x - k \\ \underline{x - k} \\ 0 \end{array}$$

$\therefore (x-k)(k^3x+1)=0$
 $x=k$ or $x = -\frac{1}{k^3}$
 Q is $(-\frac{1}{k^3}, -k^3)$

② or
 formula

method ②

product of roots

$$\alpha\beta = -\frac{k}{k^3} = -\frac{1}{k^2}$$

since $\alpha=k$ is one root

$$k\beta = -\frac{1}{k^2}$$

$$\beta = -\frac{1}{k^3}$$

$\therefore Q$ is $(-\frac{1}{k^3}, -k^3)$

1 method
 1 answer

②

③

(f) Slope $QR = \frac{-\frac{1}{k} - (-k^3)}{-k - (-\frac{1}{k^3})}$
 $= \frac{(k^4 - 1)}{\frac{-k^4 + 1}{k^3}}$
 $= -k^2$

①

Slope $PR = \frac{1}{k^2}$

Slope $QR \times \text{Slope } PR = -k^2 \times \frac{1}{k^2} = -1$

$\therefore QR \perp PR$

①

②